Phys 20BL Week7/8 Out of Lab: RCL Circuits

Zih-Yu Hsieh

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Contents

1 Physical Model for RC Circuit

1.1 Lab Guide Model

Let's start with what the lab guide says:

Given that the voltage of a capacitor is linearly proportional to the charge obtained on the capacaitor, if the proportionality constant is denoted as "capacitance" C, then it provides the following equation, with Q being the charge (unit: Coulomb C) and V being the voltage (unit: Volt V):

$$Q = CV \tag{1}$$

Also, by definition the rate of change of charge Q with respect to the time is the current of the system I (unit: Ampere A). Which, with capacitance C remaining constant, the above equation implies the following:

$$I(t) = \frac{dQ}{dt} = C\frac{dV}{dt} \tag{2}$$

Finally, using Ohm's Law, with $I = V/R$, where R is the resistance (unit: Ohms Ω). If the circuit has fixed resistance R, then one derives the following equation:

$$V(t) = I(t)R = RC\frac{dV}{dt} \tag{3}$$

Remark: This only considers the information across the capacitor, and fails to consider the general setup. Which, it can never solve the desired model on its own (as for $R, C > 0$ as assumption, this differential equation provides $V(t) = V_0 e^{t/(RC)}$, which is exponentially growing, failed to capture the general situation). So, let's include some more information.

1.2 Derivation for RC Circuits

Here, we'll fix a constant voltage source $V_s > 0$, $V(t)$ represents the voltage across the capacitor, $I(t)$ represents the total current in the circuit, C represents the constant capacitance of the capacitor, and R represents the constant resistance.

In an RC Circuit, since the voltage across the resistance and across the capacitance should sum up to the provided voltage V_0, hence one has voltage across the resistance being $V_0 - V(t)$. Also, by Ohm's Law, the voltage across the resistance is $RI(t)$, hence one has the equation $V_0 - V(t) = RI(t)$, or $RI(t) + V(t) = V_0$.

Finally, notice that $I(t) = \frac{dQ}{dt}$, where $Q(t)$ is the charge of the capacitor. Which, if $C, V(t)$ are the capacitance and voltage across the capacitor respectively, then $Q(t) = CV(t)$, hence $I(t) = \frac{dQ}{dt} = C\frac{dV}{dt}$.

Combine the two parts, we get the following differential equation:

$$RC\frac{dV}{dt} + V(t) = V_s \tag{4}$$

As a side note, for the case $V_s < 0$, this differential equation still works (just in general it's convenient to choose the direction where $V_s > 0$).

As of example, there are two functions (**Equation 1 & 2** in the In Lab portion) to consider:

$$V(t) = V_0(1 - e^{-t/\tau_{\text{step}}}) \tag{5}$$
$$V(t) = V_0^{-t/\tau_{\text{nat}}} \tag{6}$$

Let's try and solve these in terms of differential equations with initial conditions.

1.2.1 Solution for Equation 1

Given the equation $V(t) = V_0(1 - e^{-t/\tau_{\text{step}}})$, it models the circuit with the following condition:

- A fixed positive voltage source $V_s = V_0 > 0$.
- The capacitor is initially with no charge, namely $Q(0) = C \cdot V(0) = 0$. Then, it implies $V(0) = 0$.
- The resistance $R > 0$ and capacitance $C > 0$ of the circuit are fixed.

Which, solving the above differential equation under this condition, we get:

$$\begin{aligned}
&RC\frac{dV}{dt} + V(t) = V_0 && (7)\\
\implies &\frac{dV}{dt} + \frac{1}{RC}V(t) = \frac{V_0}{RC} && (8)\\
\implies &\frac{dV}{dt}e^{t/(RC)} + V(t) \cdot \frac{1}{RC}e^{t/(RC)} = \frac{V_0}{RC}e^{t/(RC)} && (9)\\
\implies &\frac{d}{dt}\left(V(t) \cdot e^{t/(RC)}\right) = \frac{V_0}{RC}e^{t/(RC)} && (10)\\
\implies &V(t) \cdot e^{t/(RC)} = V_0 e^{t/(RC)} + D && (11)\\
\implies &V(t) = V_0 + De^{-t/(RC)} && (12)
\end{aligned}$$

Under the initial condition $V(0) = 0$, one has $V_0 + D = 0$, or $D = -V_0$. Hence, the solution is given as follow:

$$V(t) = V_0\left(1 - e^{-t/(RC)}\right) \tag{13}$$

So, one has the step response time constant $\tau_{\text{step}} = RC$.

1.2.2 Solution for Equation 2

Given the equation $V(t) = V_0 e^{-t/\tau_{\text{nat}}}$. Notice that it's under the following condition:

- Zero voltage source $V_s = 0$ (modeling the situation where voltage source is turned off).
- The capacitor is fully charged initially, say $V(0) = V_0$ (normally meaning the voltage source before getting turned off).

Which, solving th above differential equation under this condition, we get:

$$\begin{aligned}
&RC\frac{dV}{dt} + V(t) = 0 && (14)\\
\implies &\frac{dV}{dt} = -\frac{1}{RC}V(t) && (15)\\
\implies &V(t) = De^{-t/(RC)} && (16)
\end{aligned}$$

Under the initial condition $V(t) = V_0$, oe has $D = V_0$. Hence, the solution is given as follow:

$$V(t) = V_0 e^{-t/(RC)} \tag{17}$$

So, one has the natural response time constant $\tau_{\text{nat}} = RC$ again.

2 Physical Model for LRC Circuit

This section is more related to the modification of experiments I've done.

Let's recall that for an inductor with fixed inductance L, when a current passes through it (assuming the current is a differentiable function), one has the following voltage change across the inductance:

$$V = -L\frac{dI}{dt} \tag{18}$$

Where, the voltage change is the voltage after the current passes through the inductor, subtracted by the voltage before the current passes through the inductor. In particular, its positive version $L\frac{dI}{dt}$ can be viewed as the voltage difference across the circuit.

2.1 Series and Parallel for Inductance

Suppose there are n inductors involved, with inductance $L_1, ..., L_n$ respectively. There are two cases to consider:

- If the inductors are in series, then notice they share a common current $I(t)$ (assume differentiability). Hence, across the ith inductor the voltage change is $V_i = -L_i\frac{dI}{dt}$, and the voltage change across the whole series arrangement is $V = \sum_{i=1}^n V_i = -\left(\sum_{i=1}^n L_i\right)\frac{dI}{dt}$.

  Which, if view the whole series arrangement with a total inductance L, then one has $V = -L\frac{dI}{dt}$, hence in general $L = \sum_{i=1}^n L_i$.

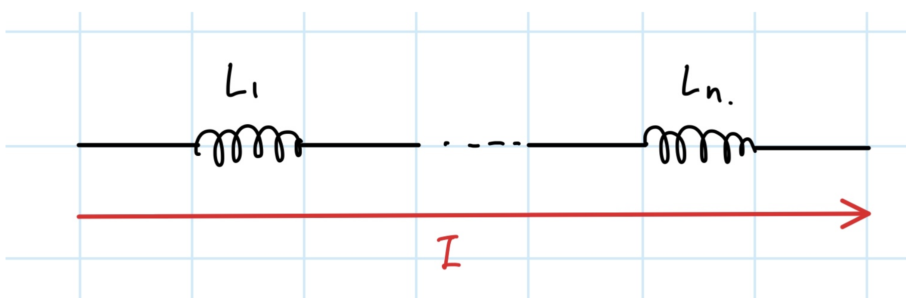


Figure 1: Inductors in Series

- If the inductors are in parallel, let I_i be the current passing through the ith inductor, then one has the total current $I = \sum_{i=1}^n I_i$. Then, notice they all share the same voltage change, say V. Hence, across the ith inductor one has voltage change $V = -L_i\frac{dI_i}{dt}$, or $-\frac{dI_i}{dt} = \frac{V}{L_i}$.

  As a result, the total current satisfies $-\frac{dI}{dt} = -\sum_{i=1}^n \frac{dI_i}{dt} = \sum_{i=1}^n \frac{V}{L_i}$, hence if L is a total inductance of the parallel arrangement, one has $V = -L\frac{dI}{dt} = V \cdot L\sum_{i=1}^n \frac{1}{L_i}$. In the general (nontrivial) situation, one has $\frac{1}{L} = \sum_{i=1}^n \frac{1}{L_i}$.

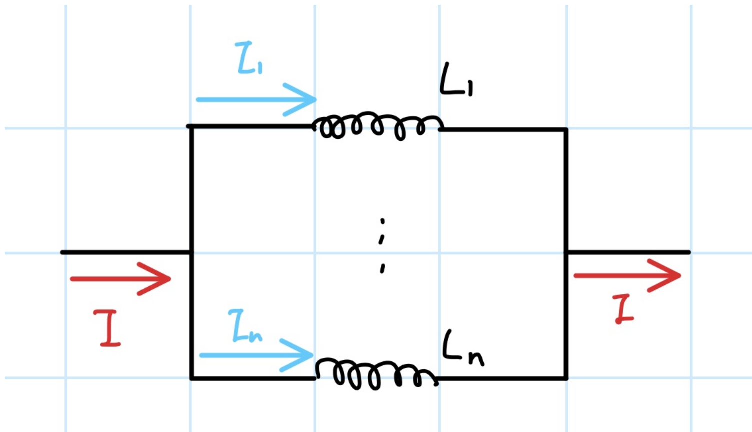


Figure 2: Inductors in Parallel

Hence, we concluded that under ideal situation, the current E&M Models predicted the total inductance to be:

- $L = \sum L_i$, given that the inductors are in series.
- $\frac{1}{L} = \sum \frac{1}{L_i}$, given that the inductors are in parallel.

2.2 Differential Equation for LRC Circuit

Now, let $V_s > 0$ represents a (fixed) voltage source, $V(t)$ be the voltage across the capacitance, L, R, C represents the inductance, resistance, and capacitance of the LRC circuit respectively. Then, with $I(t) = \frac{dQ}{dt} = C\frac{dV}{dt}$ be the current, one has $\frac{dI}{dt} = C\frac{d^2V}{dt^2}$.

Given the voltage across the inductor $L\frac{dI}{dt} = LC\frac{d^2V}{dt^2}$, the voltage across the resistance $RI(t) = RC\frac{dV}{dt}$, and the voltage across the capacitance $V(t)$, since they need to sum up to the fixed voltage source V_s, one arrives at the following differential equation:

$$LC\frac{d^2V}{dt^2} + RC\frac{dV}{dt} + V(t) = V_s \implies \frac{d^2V}{dt^2} + \frac{R}{L}\frac{dV}{dt} + \frac{1}{LC}V(t) = \frac{V_s}{LC} \tag{19}$$

Which, given the characteristic equation $r^2 + \frac{R}{L}r + \frac{1}{LC} = 0$, the roots are given by $r = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$ (which can be complex or real, dependent on the given value L). Which, labeling $w := \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$, the general solution to the homogeneous equation is:

$$V_h(t) = e^{-Rt/(2L)}\left(Ae^{wt} + Be^{-wt}\right) \tag{20}$$

Now, with the inhomogeneous part being constant $\frac{V_s}{LC}$, $V_p(t) = \frac{V_s}{LC}$ is a particular solution satisfying this differential equation, then the full solution is given as follow:

$$V(t) = V_s + e^{-Rt/(2L)}\left(Ae^{wt} + Be^{-wt}\right) \tag{21}$$

In general, the prediction of the data has $w \in i\mathbb{R}$, hence the alternative form will be withh $A\cos(|w|t) + B\sin(|w|t)$.

Finally, to obtain the voltage across the inductor, it's given as $LC\frac{d^2V}{dt^2}$, which can be solved once given all the coefficients.

2.3 Model for this Experiment

In our experiment, the provided voltage source is a square wave. However, realistically to observe the behavior of the LRC circuit, one only needs to observe the voltage within half a cycle, so one can treat the voltage source as 0 for $t < 0$, and $V_s = V_0$ for $t \geq 0$.

For simplicity of model, we'll also assume that within a short amount of time right after the voltage source is switched, the capacitance is not charged (which, it's approximated by $V(0) = 0$, or initially there's no voltage change across capacitance); on the other hand, one will also assume the voltage difference is given solely by the resistance, so $V_0 = RI(0) = RC\frac{dV}{dt}\Big|_{t=0}$. Hence, $\frac{dV}{dt}\Big|_{t=0} = \frac{V_0}{RC}$.

As for solution, notice that plugin to the general solution, one has the fllowing:

$$0 = V(0) = V_0 + A \implies A = -V_0 \tag{22}$$

$$\frac{V_0}{RC} = \frac{dV}{dt}\Big|_{t=0} = -\frac{R}{2L}A + |w|B \implies B = \frac{1}{|w|}\left(\frac{1}{RC} + \frac{R}{2L}\right)V_0 \tag{23}$$

(Note: Here we assume $w \neq 0$, as for basically 100% of the L, R, C combinations, $w \neq 0$).

3 Week 7 Portion - RC Circuit

In this part, we'll mainly be concerned with the Step responses. Which, under the assumption that capacitor is initially discharged, with the final voltage being V_0, the model is provided as $V(t) = V_0(1 - e^{-t/\tau_{\text{step}}})$, where $\tau_{\text{step}} = RC$ (under standard SI units for both R, C, then with unit of s).

3.1 Methods for Quantitative Assessment

For the Week 7 portion, as the purpose of the lab is to test the model for the voltage across the RC circuit, the methods I thought for quantitative assessment are the following (which we'll assume the data is collected over a certain bounded time interval):

Given the bounded interval, to verify the distance between the given data and the model predicted data, which there are two possible ways to approach it:

- Record the maximum difference across the interval. This is similar to the L_∞ norm in function space (which is a common norm to see the convergence of funtions, and is also a good one to compare the "closeness" of the model prediction to the actual data on certain intervals). The idea is to detect the "largest" difference between the predicted model and the actual data.

  Example: Given the the time $t_1, ..., t_n$, if $V_1, ..., V_n$ represents the collected data of the voltage, then with $V(t)$ being the provided model, then this error is given by $\max_{1 \leq i \leq n} |V(t_i) - V_i|$.

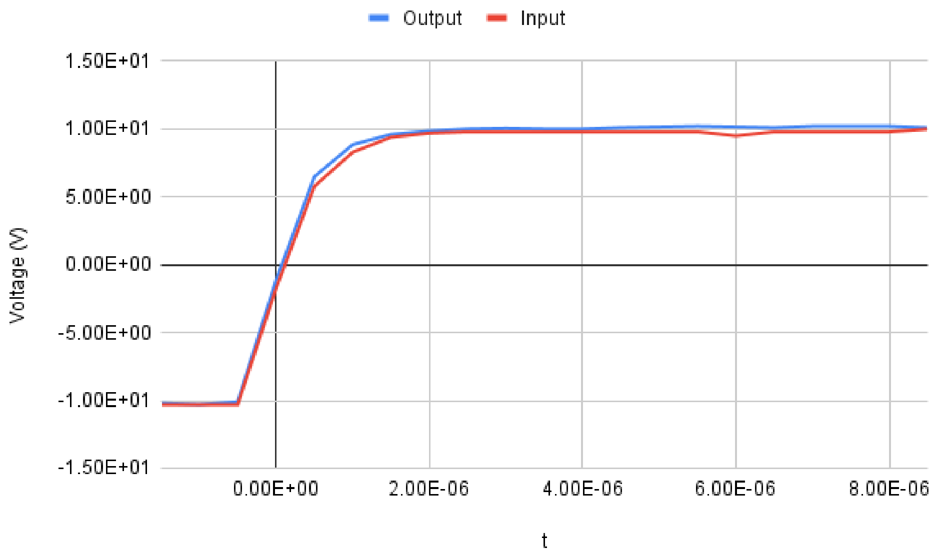
- Record the root mean square of the error. This is similar to the L_2-norm on a finite-dimensional vector space, which it views both the finite data point and the model prediction values as vectors, and takes the error as the scaled version of the L_2-norm. Which, it is more of detecting the "average" of all the errors between the data point and the predicted value.

  Example: Still, using time $t_1, ..., t_n$, data opints $V_1, ..., V_n$, and $V(t)$ as the provided model, then this error is given by $\frac{1}{\sqrt{n}} \sqrt{\sum_{i=1}^{n} (V_i - V(t_i))^2}$.

For the purpose of this lab, I'd say the root mean square of the error is a better approach, as there could have some exceptional points within the data (that drastically diverge away from the model prediction), and the maximum difference method for error would easily been affected by these exceptional points. On the other hand, the maximum difference is mostly useful when one can guarantees a bounded behavior of the error, which is not as useful in general experiments (which could have large errors compared to theoretical calculations).

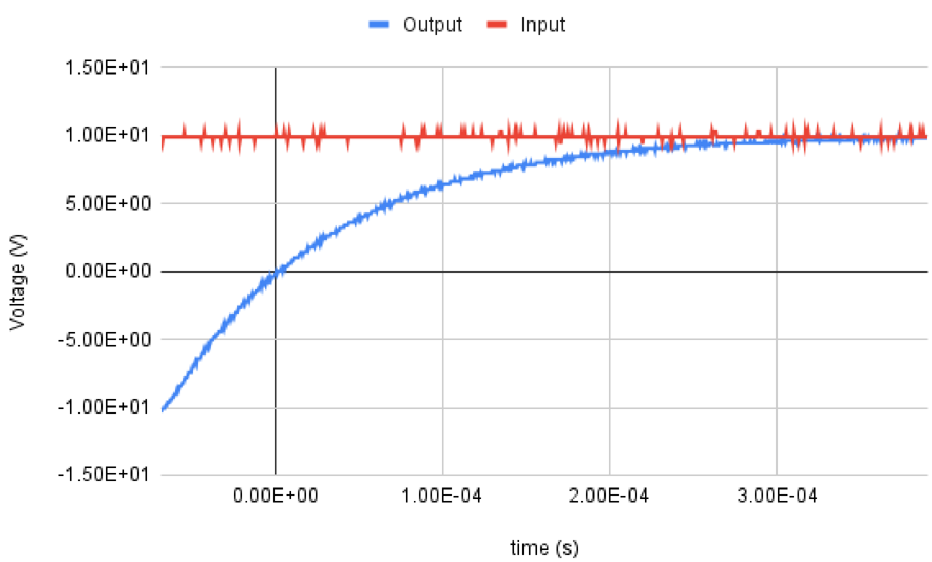
3.2 Data

Unfortunately, once the file is checked out of lab, only one out of the three trials had a distinct RC circuit response to the square wave; for the other two, the graphs of input and output voltage are nearly identical, as follow:

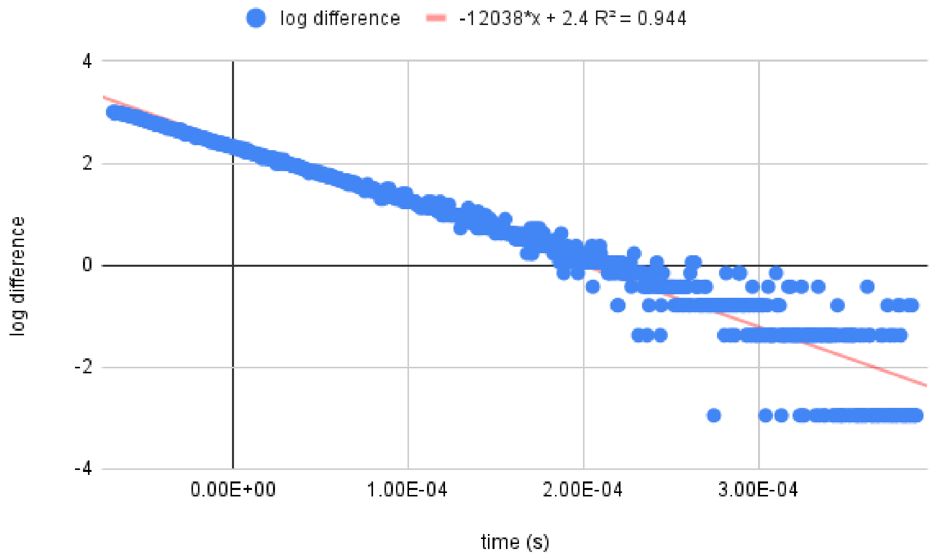


Which, actual analysis methods that should be used almost always failed here, and can't derive anymeaninful data. So, here we'll work with just 1 trial, and discuss the rest in the conclusion.

For the remaining 1 trial, the graph of the input and output voltage is given as follow (which only includes the time after the voltage is switched):



Corresponding to the voltage graph, the natural log of the difference generates the following graph:

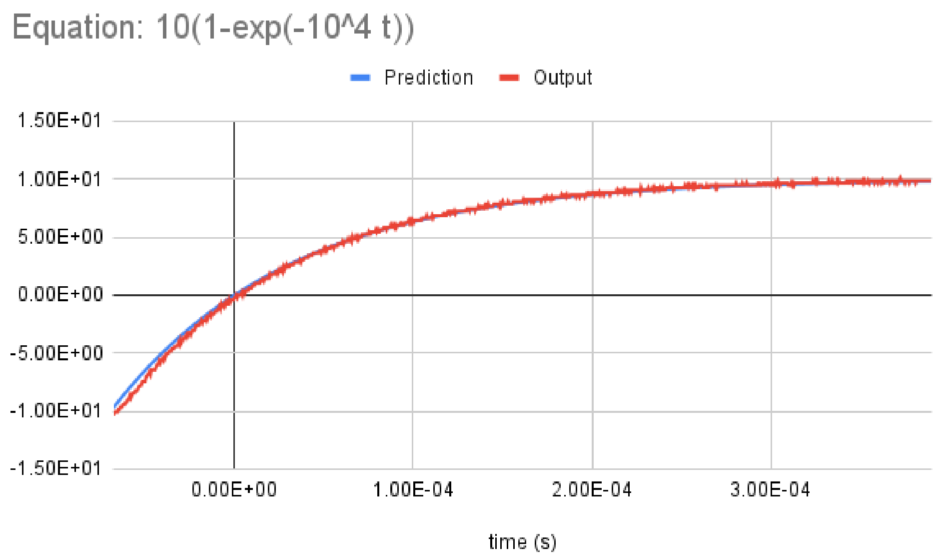


Which, it has the line of best fit $y = -12038x + 2.4$, and has slope $m = -12038$ corresponds to it.

If consider the model $V(t) = V_0(1 - e^{-t/\tau_{\text{step}}})$, then after the switch of voltage source, the input voltage is V_0, hence the difference between the input and output voltage is given by $V_0 e^{-t/\tau_{\text{step}}}$. Hence, when taking the natural log, one gets the function $\ln(V_0) - \frac{1}{\tau_{\text{step}}}t$. This shows the slope is $m = -\frac{1}{\tau_{\text{step}}}$, showing $\tau_{\text{step}} \approx 8.307 \cdot 10^{-5}$ s.

As a comparisont, for an RC circuit the step response time constance should be $\tau_{\text{step}} = RC$. Given that $R = 10^4\ \Omega$ and $C = 10^4$ pF (or 10^{-8} F), then under the SI unit the step response time constant should be 10^{-4} s. In comparison to the calculated value $8.307 \cdot 10^{-5}$ s, the percent error is about 20.380%, which is not low as of a percent error.

Finally, If plug back to the model and graph it, one faces the following graph between the output voltage and the prediction:



Which, calculating the root mean square of the error (over the 917 data points) is given by $\text{err}_{\text{rms}} \approx 0.251$ V. In comparison to the scale of 20 V in the maximum change in voltage, the root mean square error is about 1.255% of the theoretical total change, which can be considered as a great approximation.

3.3 Conclusion (Week 7)

Under the remaining 1 trial with analyzable data, the model shows a pretty good prediction of the voltage across the capacitor, with the root mean square error being 1.255% of the total change in voltage, which can be considered as a good prediction.

On the other hand, the predicted time constant is $\tau_{\text{step}} = 10^4$ s, while the best fit data shows the system has a time constant of $8.307 \cdot 10^{-5}$ s. Since there's no other good data point for comparison, the only source is the percent error, which calculated to be $2 - .38 - \%$, which is not considered low.

So, the model describes the observation very well, both qualitatively (on graph) and quantitatively (the root mean square error); however, because the lack of multiple trials with analyzable data, no other data points can be made for time constant comparison, and it diverged a bit larger away from the actual value.

For improvement, next time it's desirable to put in specific time for data check (as copying the data from oscilloscope needs to open the file and check), and the time will be worth it as it prevents situations where the datas are not analyzable.

For the actual improvement of the experiment itself (about how good the model prediction is experimentally), it'll be desirable to change the resistance and/or capacitance in the circuit, and test the time constant for each of them, then make general comparison about how these value compared to the model prediciton, and get an experimental conclusion for more combinations of RC circuits rather than just one.

4 Week 8 Portion - LRC Ciruit

For this portion, we're mainly concerned about how the series and parallel arrangement of inductors affect the total inductance. We'll calculate the predicted inductance in each case, derive the prediction of the voltage, and compare to the actual data points (using the root mean square error again).

4.1 Model Prediction

Here, the max voltage is 10 V (after the switch of square wave). Also, there are three 10^4 mH inductors that are involved. Hence, under each arrangement, here are the predicted total inductance:

- For the one in series, the total inductance $L = 3 \cdot 10^4$ mH $= 30$ H.
- For the one in parallel, the total inductance has $\frac{1}{L} = \frac{3}{10^4}$ 1/mH, hence $L = \frac{10^4}{3}$ mH $= \frac{10}{3}$ H.

Then, with fixed resistance and capacitance $R = 10^4\ \Omega$, $C = 10^4$ pF $= 10^{-8}$ F, plugin to the equations, one gets the following predicted coefficients for the model:

- For the series one, the decaying constant $\frac{R}{2L} = \frac{10^3}{6}$, while the "frequency" $w = \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} = \sqrt{\frac{10^6}{36} - \frac{10^7}{3}} \approx 1818.119i$. Also, the two coefficients have $A = -V_0 = -10$, and $B = \frac{1}{|w|}\left(\frac{1}{RC} + \frac{R}{2L}\right) V_0 \approx \frac{1}{1818.119}\left(10^4 + \frac{10^3}{6}\right) \cdot 10 \approx 55.919$, so the predicted solution is:

$$V(t) = 10 + e^{-10^3 t/6}\left(-10\cos(1818.119t) + 55.919\sin(1818.119t)\right) \tag{24}$$

  So, the voltage across the inductance is given as:

$$V_L(t) = LC\frac{d^2V}{dt^2} = e^{-10^3 t/6}\left(-0.333\cos(1818.119t) - 56.859\sin(1818.119t)\right) \tag{25}$$

- For the parallel one, the decaying constant $\frac{R}{2L} = \frac{3 \cdot 10^3}{2} = 1500$, while the "frequency" $w = \sqrt{\frac{9 \cdot 10^6}{4} - 3 \cdot 10^7} \approx 5267.827i$. Also, the two coefficients have $A = -V_0 = -10$, and $B = \frac{1}{|w|}\left(\frac{1}{RC} + \frac{R}{2L}\right) V_0 \approx \frac{1}{5267.827}\left(10^4 + 2500\right) \cdot 10 \approx 21.831$, so the predicted solution is:

$$V(t) = 10 + e^{-1500t}\left(-10\cos(5267.827t) + 21.831\sin(5267.827t)\right) \tag{26}$$

  So, the voltage across the inductance is given as:

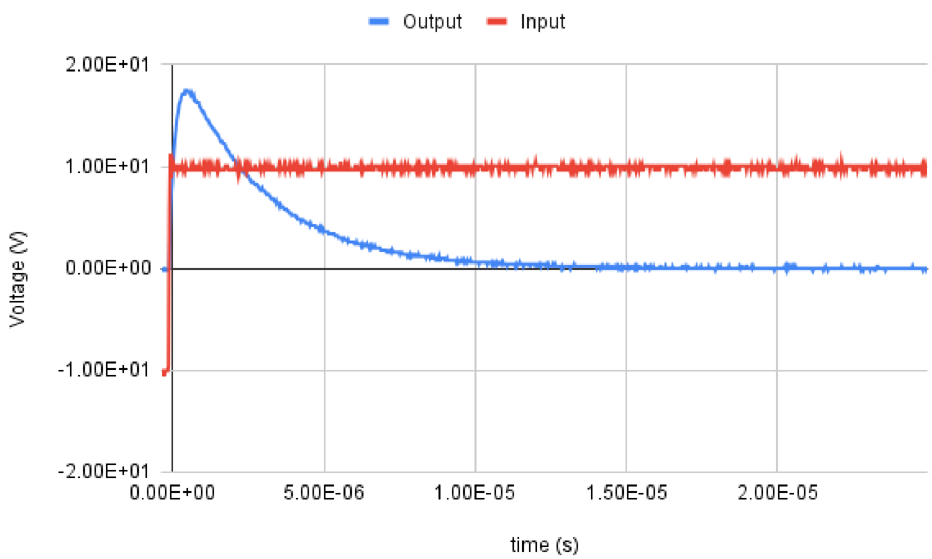
$$V_L(t) = LC\frac{d^2V}{dt^2} = e^{-1500t}\left(-3.000\cos(5267.827t) - 23.824\sin(5267.827t)\right) \tag{27}$$

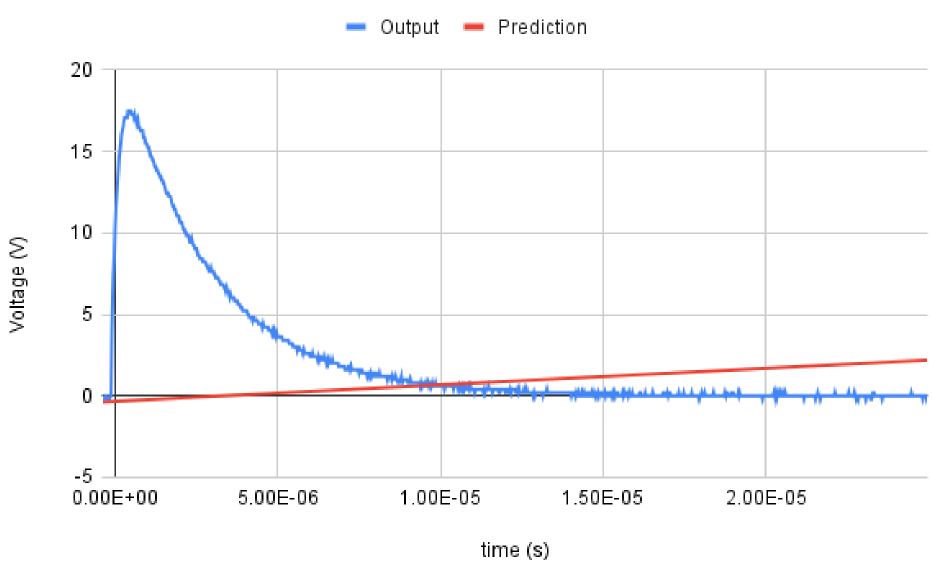
Which, it's expected that the series arrangement creates a higher inductance than the parallel arrangement. Hence, quantitatively it predicts a slower decay rate of voltage across the series of inductors, in comparison to the parallel arrangement.

On the other hand, it's also predicted that both series and parallel arrangment would have an oscillatory behavior of the voltage, with the period of series arrangement being longer than the parallel arrangement (as the frequency of parallel arrangement is predicted to be higher).

4.2 Data

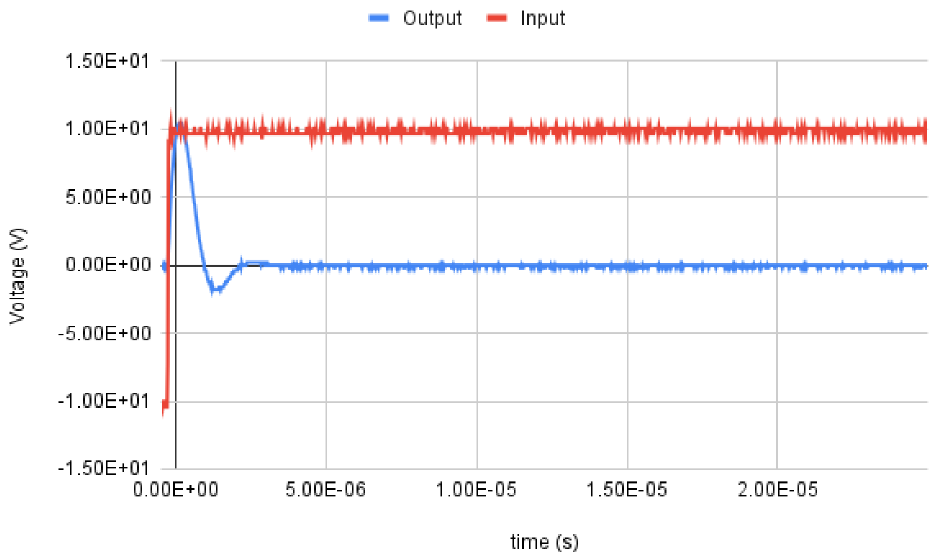
For the series arrangement, plotting the data (with input and output voltage), and another plot with the comparison of the data versus the prediction, it shows the following two:

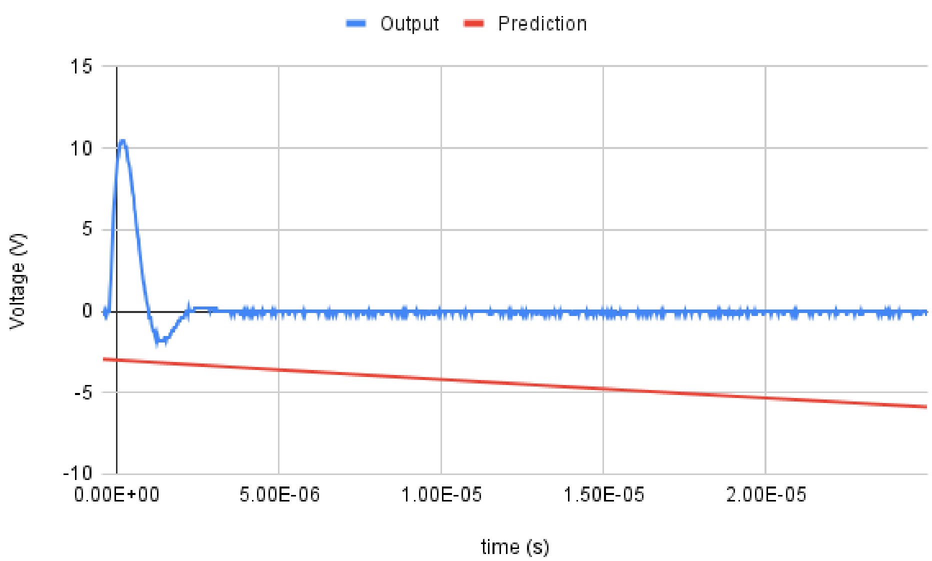




Which, the prediction diverges a lot from the actual data, with root mean quare error $\text{err}_{\text{rms}} \approx 5.011$ V, which is abot 25% of the maximum change in voltage. This concludes to be a really inaccurate prediction.

Similarly, for the parallel arrangement, it shows the following two:





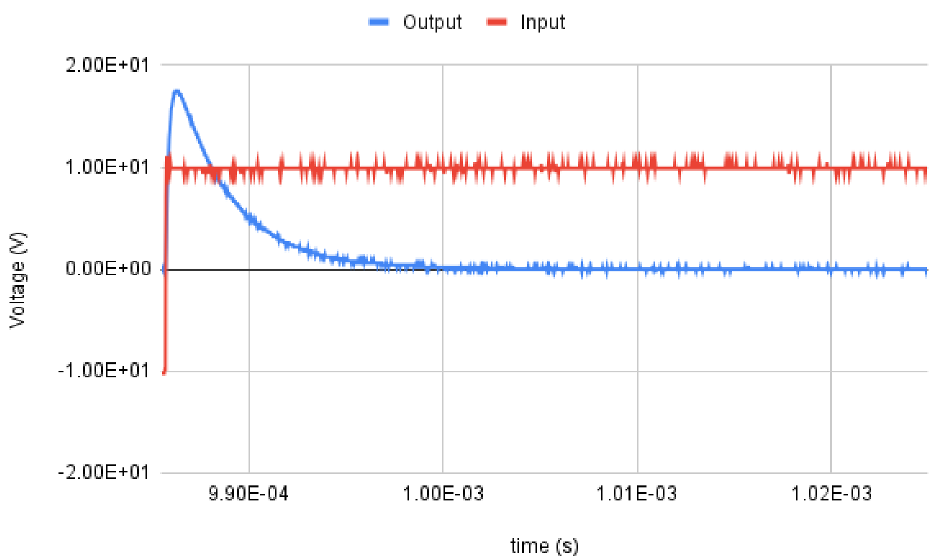
Similarly, the prediction diverges a lot from the actual data, with root mean square error $\text{err}_{\text{rms}} \approx 4.889$ V, which is also about 25% of the maximum change in voltage. This also concludes to be a really inaccurate prediction.

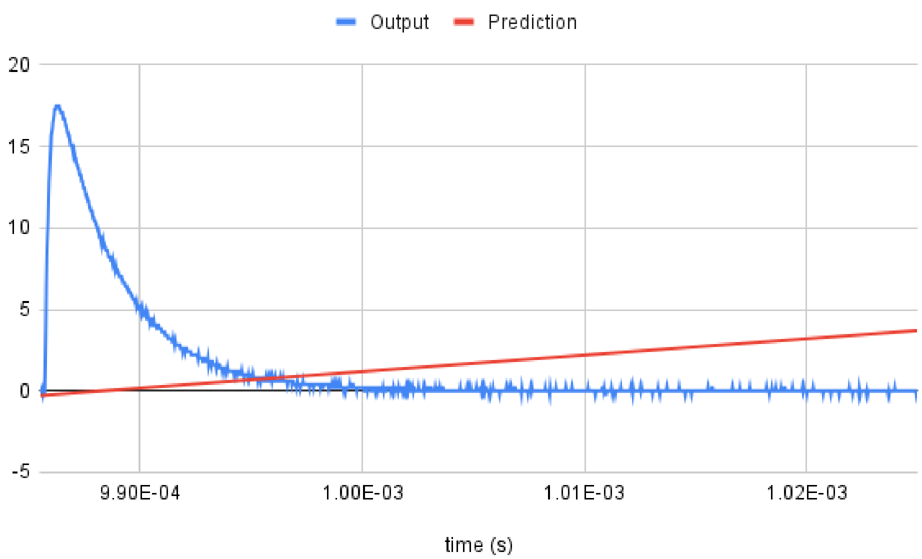
Now, if just compare the data, one can recognize that the parallel arrangement's voltage does decay a lot faster than the series arrangement, so the prediction in this part is right; on the other hand, the parallel arrangement has an oscillatory behavior, while it's not observable with naked eye for the series arrangement, hence the prediction is not as accurate for the oscillatory behavior.

For the second and the third trial, we'll simply include the data points, the predictions, and the root mean square error of the prediction (as the results are similar to trial 1).

4.3 Trial 2

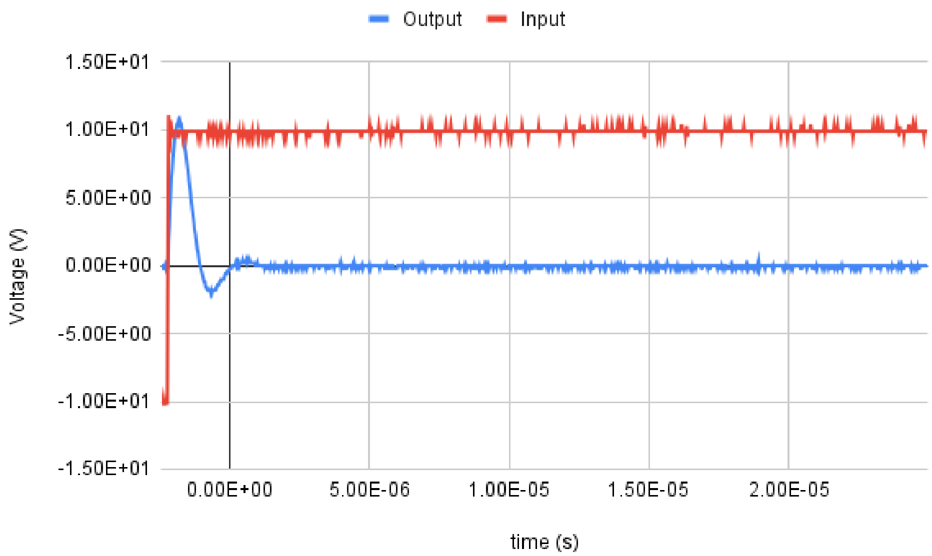
Series:

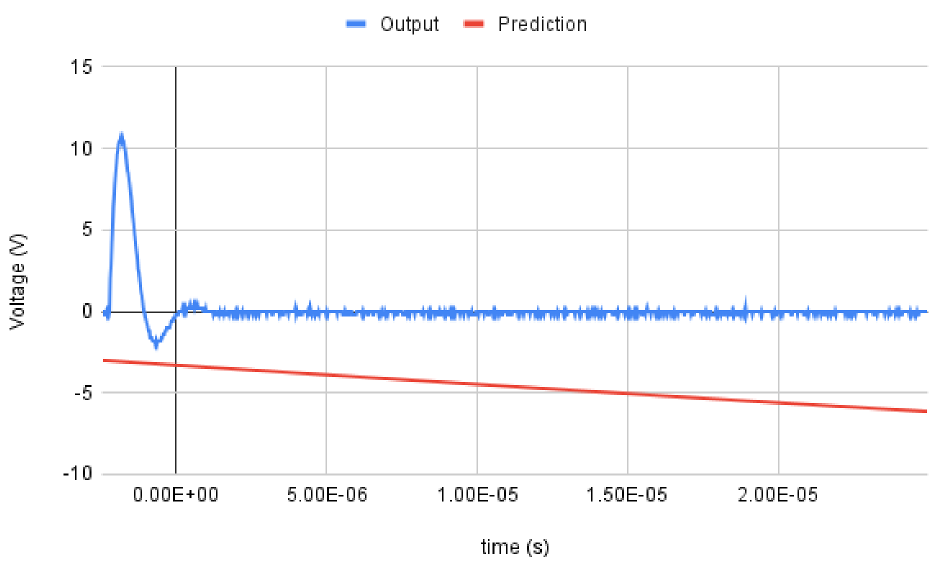




Root mean square error of prediction: $\mathrm{err}_{\mathrm{rms}} \approx 4.422$ V.

Parallel:

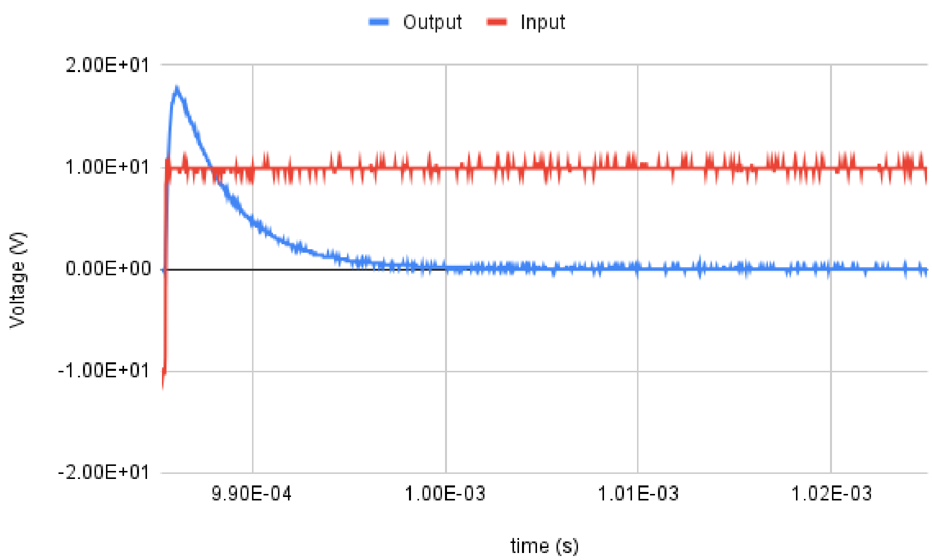


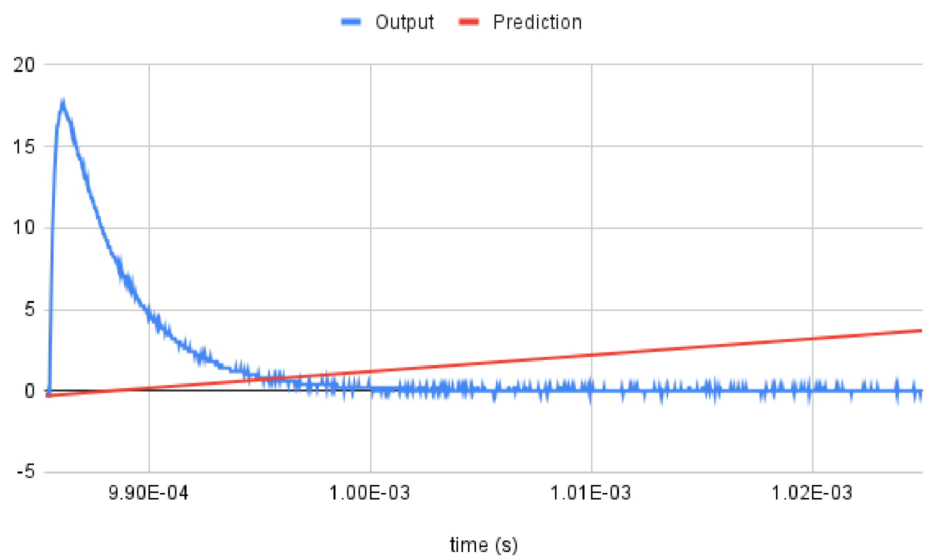


Root mean square error of prediction: $\mathrm{err}_{\mathrm{rms}} \approx 5.014$ V.

4.4 Trial 3

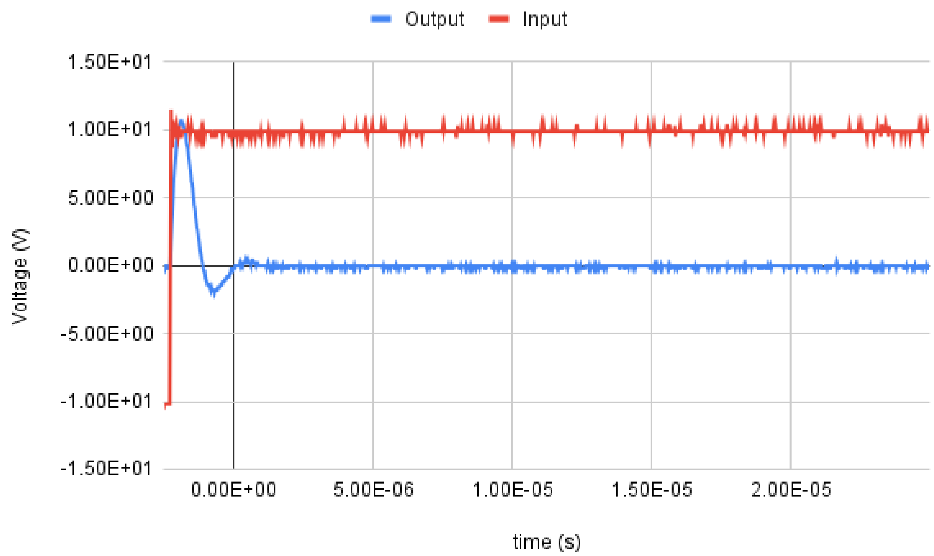
Series:

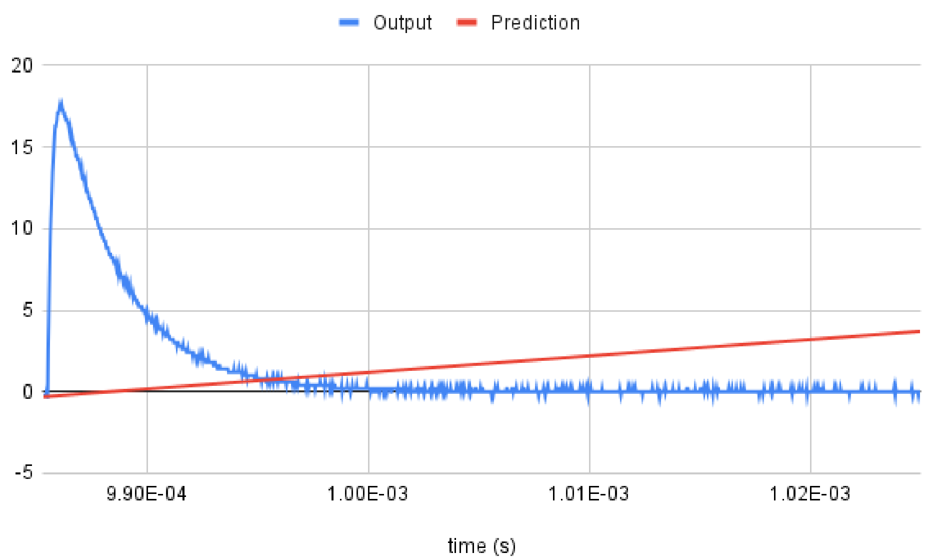




Root mean square error of prediction: $\mathrm{err}_{\mathrm{rms}} \approx 4.396$ V.

Parallel:





Root mean square error of prediction: $\mathrm{err}_{\mathrm{rms}} \approx 5.016$ V.

4.5 Conclusion